

Sixth Assignment

19.11.2024

6/6 Points

Forsøk 2



Se gjennom tilbakemelding

19.11.2024

Forsøk 2 Poengsum:

6/6



Legg til kommentar

Anonym vurdering: **nei**

3 tillatte forsøk

11.11.2024 til 20.11.2024

▼ Detaljer

Assignment: Shortest Path Detection with Negative Cycles

Problem Statement

You are given a directed graph

$G=(V,E)$ where: V is the set of vertices (nodes). E is the set of edges, some of which may have negative weights.

The graph may also contain negative weight cycles.

Your task is to:

- Implement an algorithm to find the shortest path from a given starting vertex (source) to all other vertices in the graph.
- Detect and count the number of negative weight cycles in the graph.
- This problem requires handling negative weights and detecting cycles.

Requirements

Input:

A directed graph G with vertices V and edges E , including edge weights (which may be negative).

A specific source vertex from which the shortest paths should be calculated.

The user provides an adjacency matrix for a graph (get the adjacency matrix from user input).

Use INF to represent no direct edge and 0 to indicate no self-loop.

Output:

- The shortest path from the source to each reachable vertex.
- The total number of negative weight cycles present in the graph.

For vertices within a negative cycle, indicate that they are "unreachable" from the source due to the cycle.

Algorithm: Use the Bellman-Ford algorithm, which is suitable for graphs with negative weights

and capable of detecting negative weight cycles.

Edge Cases:

Ensure your solution handles cases where

- The graph has one or more negative weight cycles.
- Some vertices are unreachable from the source.
- The graph is disconnected.

Scoring:

- Correctly implementing the Bellman-Ford algorithm to find the shortest paths from the source vertex, adjacency matrix user input. **(3 points)**
- Accurately detecting and counting the number of negative weight cycles in the graph. **(2 points)**
- Handling edge cases where vertices are unreachable or affected by negative cycles. **(1 point)**

Submission:

Submit your assignment in a ZIP file.
